

Microeconomics

Lecture 19: Asymmetric Information

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What We Know

In the models studied so far we have assumed **perfect information**:

- Perfect competition: everyone knows the prices, quality, and characteristics of every good
- Monopoly: the firm knows demand; consumers know the price
- Oligopoly: firms know each other and know their cost functions

We have seen that **market power creates a deadweight loss (DWL)**: the monopolist or the cartel produces *too little* and sells at prices that are *too high* relative to the social optimum.

Today we address a distinct and often more pervasive source of inefficiency. Information is almost always asymmetric. One party knows things the other does not:

- The seller of a used car knows whether it is defective; the buyer does not
- The worker knows how productive she is; the employer does not
- The patient knows whether he is healthy or sick; the insurer does not
- The borrower knows whether she will be able to repay; the bank does not

Focus of the lecture

Asymmetric information is a **market failure**: it distorts trade and sometimes eliminates it altogether. We study the two main types, their effects, and the private and public mechanisms that mitigate them.

Learning Objectives

1. **Distinguish** hidden information (adverse selection) from hidden action (moral hazard)
2. **Explain** Akerlof's model and the mechanism of adverse selection
3. **Describe** how adverse selection can lead to market collapse and apply it to real cases (insurance, credit)
4. **Define** moral hazard and illustrate it with a numerical model
5. **Recognize** the remedies: signaling, screening, reputation, regulation
6. **Construct** a separating equilibrium in Spence's signaling model
7. **Solve** exercises on adverse selection, signaling, and moral hazard

Common thread

Asymmetric information is one of the main market failures studied in microeconomics. Understanding it is essential for many policy areas: banking regulation, the health care system, the labor market, consumer protection.

Two Types of Asymmetric Information

Hidden information

One party knows **characteristics** that the other does not observe *before* the transaction.

Problem: *adverse selection*

Examples: quality of a used car, health status of the insured, credit risk of the borrower, productivity of the worker.

Hidden action

One party takes **actions** that the other does not observe *after* the contract is signed.

Problem: *moral hazard*

Examples: the insured drives more riskily, the employee works less if not monitored, banks take on risk if they know they are “too big to fail”, the doctor prescribes unnecessary treatments paid for by the national health service (SSN).

In both cases: the uninformed party (the *principal* in contract theory) cannot observe everything it needs to make optimal decisions.

Akerlof's Model: The Market for "Lemons"

George Akerlof (Nobel Prize 2001) analyzed the used-car market in his seminal article "The Market for Lemons" (1970).

Model setup:

- Two types of used cars: *good* ("peaches") and *bad* ("lemons")
- **Sellers** know what type of car they have
- **Buyers** cannot tell them apart before purchase

Subjective valuations:

	Good car	Bad car
Value to the buyer	€12,000	€6,000
Value to the seller	€10,000	€4,000

Note. There are always gains from trade: the buyer values each type of car more than the seller does. With perfect information, *all* types of cars would be traded.

Akerlof: The Problem of Asymmetric Information

Suppose the share of good cars in the market is p and that of bad cars is $(1 - p)$.

The buyer does not observe the type, so she offers the expected price:

$$WTP_{\text{buyer}} = p \cdot 12,000 + (1 - p) \cdot 6,000 = 6,000 + 6,000 p$$

The seller of a good car sells only if:

$$\text{Price offered} \geq 10,000 \implies 6,000 + 6,000 p \geq 10,000 \implies p \geq \frac{2}{3}$$

What happens if $p < 2/3$?

Sellers of good cars do not sell (the price offered does not cover their reservation value). Only bad cars remain on the market. But then the actual $p \rightarrow 0$: the buyer offers only €6,000 (the expected value of a bad car). The market for good cars progressively collapses.

Akerlof: The Mechanism of Adverse Selection

The mechanism in three steps:

1. **The buyer does not observe the type** $\Rightarrow$ offers an average price based on the expected share of good and bad cars
2. **Sellers of good cars withdraw** because the average price is too low (it does not cover their reservation value)
3. **The share of good cars falls** $\Rightarrow$ the buyer lowers the offer $\Rightarrow$ even more sellers of good cars withdraw $\Rightarrow$ downward spiral

Adverse selection

Adverse selection is the phenomenon whereby, under asymmetric information, **low-quality** products tend to prevail in the market, driving out high-quality ones. The market “selects” the worst types.

Result. Mutually beneficial trades (the good cars) do not take place. This is an *efficiency loss* caused by asymmetric information, not by wrong prices.

Adverse Selection: Example

Data: $p = 1/2$ (half good cars, half bad).

Step 1. Buyer offers: $WTP = 6,000 + 6,000 \cdot \frac{1}{2} = 9,000$.

Step 2. Do sellers of good cars sell, i.e., is $9,000 \geq 10,000$? *No*. They withdraw.

Step 3. Only bad cars remain: $p \rightarrow 0$.

Step 4. Buyer offers: $WTP = 6,000 + 0 = 6,000$.

Step 5. Do sellers of bad cars sell, i.e., is $6,000 \geq 4,000$? *Yes*.

Final equilibrium

Only **bad** cars are traded, at €6,000. Good cars remain **unsold**, even though gains from trade exist (€12,000 buyer > €10,000 seller). The market for good cars has **collapsed**. If $p \geq 2/3$: both types are traded.

Adverse Selection: The Health Insurance Market

The insurance market is the most important and most studied case of adverse selection.

Setup. An insurance company does not know the health status of applicants. It offers a single premium based on the average risk of the population.

Who buys insurance?

- **Healthy** people know they are low-risk. They find the premium expensive and tend not to buy insurance.
- **Sick or high-risk** people find the premium attractive and buy insurance en masse.

Consequence: the pool of the insured is worse than average $\Rightarrow$ the company raises the premium $\Rightarrow$ even more healthy people drop out $\Rightarrow$ upward spiral of premiums.

Failure of the private market

The private insurance market tends to *exclude* those most in need (the chronically ill, the elderly, people with pre-existing conditions) and to be undersized relative to the social optimum. This is one of the main arguments for compulsory public health insurance.

Adverse Selection: The Credit Market

Banks and adverse selection.

Problem. The bank knows neither the quality of the applicant's project nor her ability to repay. It sets an interest rate r based on average risk.

Effect of adverse selection:

- Applicants with **safe projects** (low return, low risk) find the rate too high and give up on credit
- Applicants with **risky projects** (high return if successful but high probability of failure) find the rate attractive and borrow
- The bank ends up with a portfolio overpopulated by risky borrowers

Implication: credit rationing (Stiglitz & Weiss, 1981)

The bank *cannot* simply raise the rate to compensate for risk: raising r further worsens the composition of the applicant pool (even more adverse selection). The equilibrium solution is often to **ration credit**: not lending to certain applicants even if they would be willing to pay higher rates.

Business Case: How Digital Platforms Respond to Akerlof (1/2)

Original problem (Akerlof, 1970). In one-shot markets between private parties (cars, real estate, services) quality is not observable: high-quality sellers exit the market.

AutoScout24, Autovita, CarGurus—reducing information asymmetry through data

Digital platforms fight adverse selection by aggregating information:

- **Vehicle history** (certified mileage, number of owners, accidents from Carfax/AutoDNA)
- **Seller rating**: observable reputation curbs opportunistic behavior
- **On-demand inspection**: on AutoScout24 the buyer can request an ACI/RAC inspection before purchase

Business Case: How Digital Platforms Respond to Akerlof (2/2)

Certified Pre-Owned (CPO) programs—the warranty as a costly signal

BMW, Mercedes, Toyota, Volkswagen offer used cars with a 1–2 year official warranty. CPO cars are selected, inspected, and reconditioned by the manufacturer.

- The warranty is a **credible and costly signal**: only a genuinely good car can sustain it without losses
- Average price premium relative to private sales: €1,500–3,500 (same car, same year, same mileage)

Result: technology reduces—but does not eliminate—the asymmetry. The market for good cars revives, at the cost of intermediation.

In Depth: Rothschild and Stiglitz (1976)

Fundamental question: Can private insurance markets reach an efficient equilibrium when consumers' types are not observable?

The model.

- Two types of insured: **low risk** (type L , accident prob. p_L) and **high risk** (type H , prob. p_H), with $p_H > p_L$
- The insured knows her own type; the company does not
- Loss L if an accident occurs, 0 otherwise; initial wealth W_0
- Competing insurance companies offer contracts (q, K) : premium q , indemnity K
- The insured maximize expected utility with concave U (risk averse)

State of the world $\rightarrow$ contract

Contract $(q, K) \Rightarrow$ wealth in the no-accident state: $W_1 = W_0 - q$; in the accident state: $W_2 = W_0 - L + K - q$. The (W_1, W_2) plane is the natural space for analyzing insurance contracts.

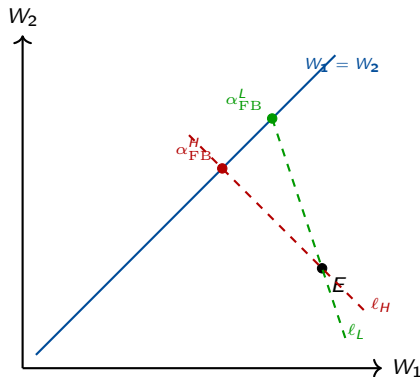
RS: The (W_1, W_2) Plane and the Fair-Odds Lines

Reference points in the (W_1, W_2) plane:

- $E = (W_0, W_0 - L)$: endowment without insurance (below the diagonal)
- **45° diagonal**: $W_1 = W_2$, full insurance
- **Fair-odds line** for type p : actuarially fair contracts, $q = p \cdot K$; slope $= -(1 - p)/p$

Preferences: indifference curves in (W_1, W_2) are convex; the MRS on the diagonal is $-(1 - p)/p$. Types with higher p (type H) have flatter curves.

Optimal insurance for a type with risk p at fair odds: **full insurance**—the point at the intersection of the fair-odds line and the diagonal.



RS: First Best and Why Pooling Fails

First best (observable types)

The company offers each type a separate contract on its own fair-odds line. Since the insured are risk averse, the optimum is **full insurance**:

$$\alpha_{\text{FB}}^t = (W_0 - p_t L, W_0 - p_t L), \quad t \in \{H, L\}$$

Neither type bears any residual risk; H pays more.

Pooling equilibrium: unstable

With unobservable types, a company offers a single contract at the average fair odds

$\bar{p} = \lambda p_H + (1 - \lambda)p_L$. Type- L agents are overcharged:

1. A competitor offers slightly less coverage at a unit premium below the pooling one (but above L 's fair odds)
2. Only type- L agents accept (*cream-skimming*)
3. The competitor earns profit $> 0 \Rightarrow$ contradiction

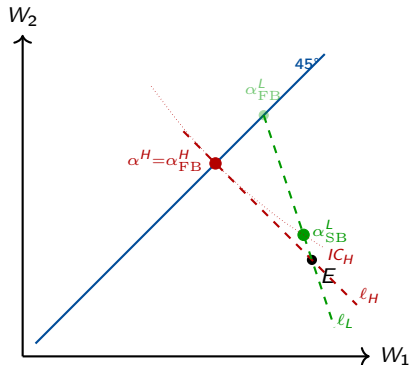
No pooling equilibrium survives competition.

RS: The Separating Equilibrium

Structure of the separating equilibrium (if it exists):

- Type H : **full insurance** at H 's fair odds, $\alpha^H = \alpha_{FB}^H$
- Type L : contract α^L on L 's fair-odds line, but with **partial coverage** (below the diagonal)
- α^L is chosen at the point on ℓ_L where H is *indifferent* between α^H and α^L — H has no incentive to mimic L

The (counterintuitive) logic. H values coverage more at the margin (high accident probability): she would like to mimic L to get full insurance at favorable fair odds. The only way to deter her is to *reduce* coverage in α^L —residual risk weighs more on H than on L . The cost of asymmetric information thus falls on the *low-risk* types.



RS: Welfare, Non-Existence, and Policy

Welfare costs (counterintuitive result)

- Type H : gets $\alpha^H = \alpha_{FB}^H$ (full insurance)—no distortion
- Type L : underinsured relative to the first best (α_{SB}^L below the diagonal)—pays the “price” of asymmetric information
- It is the *low-risk* types who bear the cost: they must “signal” themselves by giving up part of the coverage to deter H from mimicking them

The non-existence result

If the share of type- H agents is *low enough*, a pooling contract would be preferred by both types to the separating menu, but it too is unstable. The competitive market **has no equilibrium at all**—a unique result in economics.

Policy implications

- **Compulsory insurance** (SSN, INPS): imposes pooling “from above”, eliminates cream-skimming by including all types by law
- **Community rating** (ACA 2010, USA): prohibits differentiating premiums by health status or pre-existing conditions
- **Insurance mandate** (Obamacare, compulsory car liability insurance): forces type- L agents into the pool too, preventing them from leaving and leaving only type- H agents

Moral Hazard: Definition and Structure

Moral hazard

Moral hazard arises when one party, after signing a contract, has an incentive to take actions that increase the probability of adverse events borne by the other party, because she is *protected* from the consequences.

Structure of the problem:

1. **Before the contract:** the parties agree on terms that provide coverage
2. **After the contract:** the insured party changes behavior (reduces effort, increases risk) because she does not bear the consequences
3. **The other party does not observe** the action (or observes it only at high cost)

Key difference from adverse selection

- Adverse selection: information hidden *before* the contract (who you are)
- Moral hazard: action hidden *after* the contract (what you do)

Moral Hazard: Examples

1. **Car insurance:** Before being insured, the driver is careful. With full coverage, the incentive to be careful weakens: she parks less carefully, drives faster. The insurer does not observe every single action.
2. **Labor market (principal-agent):** The employer wants the employee to work hard; the employee prefers to reduce effort (it is costly); the employer cannot monitor everything. How should the contract be designed to align incentives?
3. **“Too big to fail”:** Large banks know the government will bail them out in a crisis: they take excessive risks knowing that losses will be socialized. The 2008 crisis: a textbook case of systemic moral hazard.
4. **Health care system:** Once covered by the SSN, patients may demand more treatment than necessary; doctors, paid per service, may prescribe unnecessary tests. Moral hazard is two-sided.

Business Case: AIG and “Too Big to Fail” (2008)

Context. American International Group (AIG) was the largest insurance company in the world. Between 2003 and 2007, its AIG Financial Products division issued *credit default swaps* (CDS) worth **441 billion dollars** on bonds backed by subprime mortgages (MBS/CDO).

Where moral hazard arises

- AIG knew it was *systemically important*: in a crisis, the government could not let it fail without bringing down the entire financial system
- This belief reduced the incentive to price risk correctly and to hold adequate reserves
- AIG's internal models (based on VaR) systematically underestimated default correlations

The bailout and the regulatory response

- September 2008: the Federal Reserve steps in with an emergency loan of **\$85 billion**; the total bailout would reach **\$182 billion**
- **Basel III (2010)**: stricter capital requirements, leverage ratio, liquidity coverage ratio
- **Dodd-Frank Act (2010)**: mandatory central clearing for OTC derivatives $\Rightarrow$ *skin in the game* for large banks

Moral Hazard in Insurance: A Model

Numerical example. Accident probability depends on precaution; damage: €1,000.

- High precaution (personal cost €100): accident prob. = 10%
- Low precaution (cost €0): accident prob. = 40%

Total expected cost (uninsured individual):

- High: $0.10 \cdot 1,000 + 100 = 200$ Low: $0.40 \cdot 1,000 + 0 = 400$

Without insurance: chooses high precaution ($200 < 400$). **With full insurance:** the damage is covered, so she chooses *low* precaution to save €100.

Consequence

The insurer anticipates post-contract behavior and raises the premium. If it rises too much, the trade does not take place. Moral hazard reduces the benefits of insurance and in extreme cases can eliminate them.

In Depth: The Formal Principal-Agent Model

Structure of the model

- **Principal** (firm): risk neutral, maximizes $E[\pi - w]$
- **Agent** (manager/employee): risk averse, maximizes $E[U(w)] - c(e)$, with $U'' < 0$
- Effort $e \in \{e_L, e_H\}$; cost $c_H > c_L = 0$
- Output $\pi \in \{\pi_H, \pi_L\}$: prob. p_H if e_H , prob. $p_L < p_H$ if e_L
- Agent's reservation utility: $\bar{u}$ (outside option)

The principal's two constraints:

- **Participation Constraint (PC)**: the agent must prefer the contract to the outside option:

$$p_H U(w_H) + (1 - p_H) U(w_L) - c_H \geq \bar{u}$$

- **Incentive Constraint (IC)**: the agent must prefer e_H to e_L :

$$\underbrace{(p_H - p_L)}_{\Delta p} [U(w_H) - U(w_L)] \geq c_H - c_L \equiv \Delta c$$

PA: First Best—Observable Effort

If the principal observes and contracts directly on effort:

- The IC is irrelevant: the contract specifies $e = e_H$ directly
- Risk-neutral principal, risk-averse agent: the optimal solution provides **full insurance** for the agent, i.e., a fixed wage w^* independent of output
- The PC holds with equality (the principal extracts all the surplus):

$$U(w^*) - c_H = \bar{u} \Rightarrow w^* = U^{-1}(\bar{u} + c_H)$$

Example (first best)

$U(w) = \sqrt{w}$, $\bar{u} = 4$, $c_H = 2$, $p_H = 0.75$, $p_L = 0.25$, $\pi_H = 100$, $\pi_L = 20$.

$$\sqrt{w^*} = \bar{u} + c_H = 4 + 2 = 6 \Rightarrow \boxed{w^* = 36}$$

$$E[\pi \mid e_H] = 0.75 \cdot 100 + 0.25 \cdot 20 = 80 \quad \Pi^{FB} = 80 - 36 = 44$$

The agent bears *zero* risk; the principal bears all the output risk.

PA: Second Best—The Moral Hazard Problem

If the principal does *not* observe effort:

The first-best contract induces e_L

With $w_H = w_L = 36$, the agent chooses the effort that maximizes $U(36) - c = 6 - c$:

- e_H : payoff = $6 - 2 = 4 = \bar{u}$ (indifferent)
- e_L : payoff = $6 - 0 = 6 > \bar{u}$ (**prefers e_L**)

The fixed contract is not incentive compatible: moral hazard.

Inducing e_H requires a wage *spread* $w_H > w_L$:

$$\text{IC (binding): } \Delta p \cdot [\sqrt{w_H} - \sqrt{w_L}] = \Delta c \Rightarrow 0.5(\sqrt{w_H} - \sqrt{w_L}) = 2$$

$$\text{PC (binding): } 0.75\sqrt{w_H} + 0.25\sqrt{w_L} = \bar{u} + c_H = 6$$

PA: Solution and Comparison

Solving the IC + PC system:

$$\sqrt{w_H} = \sqrt{w_L} + 4 \xrightarrow{\text{subst. in PC}} 0.75(\sqrt{w_L} + 4) + 0.25 \sqrt{w_L} = 6 \Rightarrow \sqrt{w_L} = 3 \Rightarrow \boxed{w_L = 9}$$

$$\sqrt{w_H} = 7 \Rightarrow \boxed{w_H = 49}$$

Comparison of first best vs second best:

	First best	Second best
w_H (wage if success)	36	49
w_L (wage if failure)	36	9
Expected wage $E[w]$	36	$0.75 \cdot 49 + 0.25 \cdot 9 = 39$
Principal's profit Π	$80 - 36 = 44$	$80 - 39 = 41$
Agency cost (risk premium)	0	$39 - 36 = 3$
Risk borne by the agent	0	$w_H - w_L = 40$

PA: The Insurance-Incentives Trade-off

Central result of the principal-agent model

Inducing e_H requires exposing the agent to risk (wage spread). The agent is risk averse and must be compensated: the principal must pay a **risk premium** of 3 (the agent stays at $\bar{u}$). Asymmetric information has a real and direct cost.

The linear contract. Many applications use $w = \alpha + \beta \pi$:

- $\beta = 0$: fixed wage, full insurance, no incentive $\Rightarrow e_L$
- $\beta = 1$: the agent “buys the firm”, maximum incentives, maximum risk
- Optimum: $\beta^* \in (0, 1)$, increasing in Δc , decreasing in Δp and in risk aversion

Applications

- **Executive compensation**: fixed salary + bonus + stock options—high β for CEOs
- **Sales forces**: almost entirely on commission ($\beta \approx 1$)
- **Pre-2008 bank bonuses**: unlimited upside, zero downside $\Rightarrow$ asymmetric β that encourages excessive risk-taking—systemic moral hazard
- **CRD IV (EU)**: variable $\leq 100\%$ of fixed (200% if shareholders approve), *clawback*, 3–5 year deferral

Remedies for Asymmetric Information

How can the distortions caused by asymmetric information be mitigated?

Against adverse selection:

1. **Signaling:** the informed party sends a credible signal about quality
2. **Screening:** the uninformed party offers a self-selecting menu
3. **Reputation:** observable history of quality
4. **Certifications and warranties:** third parties
5. **Regulation:** compulsory insurance, disclosure

Against moral hazard:

1. **Monitoring:** observing actions (costly)
2. **Performance incentives:** bonuses, commissions, stock options
3. **Deductibles/co-payments:** “skin in the game”
4. **Long-term contracts and reputation**
5. **Regulation:** prudential rules, capital requirements (Basel III)

Signaling: Spence's Model

Michael Spence (Nobel Prize 2001 with Akerlof and Stiglitz) developed the model of **signaling in the labor market**.

Setup:

- Two types of workers: high productivity (H) and low productivity (L)
- Workers know their own type; employers do not
- Employers offer wages based on expected productivity

The signal. The worker can acquire **education** (a degree, certifications). Education:

- Has a *cost* for workers (time, effort, money)
- The cost is **lower for type- H workers** (they learn with less effort)

Signaling works even if education does not raise productivity

Spence's key idea is radical: education can be a credible signal *even if it teaches nothing useful for the job*, as long as it is more costly for type L than for type H . Type- L workers do not want to mimic it because the cost exceeds the wage benefit.

Signaling: Separating Equilibrium

Numerical example.

- Productivity of H : €80,000/year. Productivity of L : €50,000/year.
- Cost of education (degree) for type H : €20,000 (in present value)
- Cost of education for type L : €40,000

Candidate separating equilibrium. Type- H workers get a degree and earn $w_H = 80,000$; type- L workers do not get a degree and earn $w_L = 50,000$.

Checking the incentive-compatibility conditions:

$$H : 80,000 - 20,000 = 60,000 > 50,000 \quad \checkmark \text{ (worth getting a degree)}$$

$$L : 80,000 - 40,000 = 40,000 < 50,000 \quad \checkmark \text{ (not worth mimicking)}$$

Result and implication

The degree *separates* the two types: no one has an incentive to deviate. **Implication:** part of education spending is “waste” from a productivity standpoint—it serves only to signal the type. It is a social cost of asymmetric information, not an investment in human capital.

Screening: The Uninformed Party Moves First

Screening

Screening is a mechanism by which the *uninformed* party designs a menu of contracts that induces the informed party to **self-select**, indirectly revealing its type.

Difference from signaling: in signaling the informed party moves first (the worker decides whether to get a degree). In screening the uninformed party moves (the insurer designs the menu).

Example: insurance with a self-selecting menu

A company does not know whether an applicant is high- or low-risk. It offers two policies:

- Policy A: low premium, high deductible (suited to low-risk types)
- Policy B: high premium, zero deductible (suited to high-risk types)

If the parameters are well calibrated, low-risk types choose A and high-risk types choose B. The company separates the types without observing them directly. Other examples: menus of employment contracts with different fixed/variable pay mixes, basic vs premium smartphone plans, optional warranties.

Business Case: The Black Box in Car Insurance (UnipolSai, Generali)

Context. Traditional car liability insurance suffers from *both* information problems: adverse selection (the company does not know the driver's risk profile) and moral hazard (once insured, the driver may drive more recklessly).

How insurance telematics works (telematics / UBI)

A GPS/accelerometer device installed in the car records in real time: speed, sudden braking, acceleration, time of driving (night/day), km driven.

- **Screening function:** the observed driving profile allows the premium to be based on actual risk rather than average risk $\Rightarrow$ safe drivers do not subsidize dangerous ones
- **Anti-moral-hazard function:** the driver knows she is monitored $\Rightarrow$ keeps a careful driving style even after signing

Italy is the most advanced market in Europe

According to IVASS data (IPER survey, 2023–2024), about **18%** of Italian car liability contracts for passenger cars include a black box—one of the highest shares in Europe. Main providers include UnipolSai, Generali, Direct Line. “Virtuous” drivers get discounts of up to 30% on the standard premium.

Reputation and Warranties as a Solution

In repeated markets, **reputation** can substitute for direct information about quality.

Mechanism. A high-quality firm invests in reputation (advertising, warranties, brand). If it cheats customers, it loses its reputation and the future value of the brand: the cost of losing reputation is high enough to make honest behavior worthwhile.

Examples of reputation as a signal

- **Amazon, Airbnb:** review systems that reduce information asymmetry about sellers/hosts
- **Pharmaceutical brands** (Bayer, Johnson & Johnson): price premium for a reputation for reliability
- **Multi-year warranties:** a 5-year warranty is a credible signal of quality—a poor producer could not afford it because it would pay out too many claims

Limitation. Reputation works in *repeated* markets. In *one-shot* transactions (selling a house, a used car between private parties) the mechanism is weaker, and we are back to Akerlof's problem.

The Role of Public Regulation

When private mechanisms (signaling, screening, reputation) are insufficient, **public regulation** can correct information asymmetries.

Against adverse selection:

- **Compulsory insurance:** car liability, INPS.
Extends the pool to all types, avoiding the adverse selection spiral.
- **Disclosure requirements:** ingredients, side effects, credit ratings, financial prospectuses.
- **Quality standards:** safety and hygiene rules, certifications.

Against moral hazard:

- **Skin in the game:** funds must retain a share of the risk
- **Capital requirements:** bank equity capital (Basel III)
- **Inspections and controls:** food producers, construction sites, health facilities
- **Legal liability:** whoever causes damage is financially liable for it

Caveat

Regulation is not a magic wand: it can create *new* moral hazard problems (state guarantees encourage risky behavior; compulsory insurance can induce overuse). Cost-benefit analysis is unavoidable.

Exercise 1—Adverse Selection: Car Market

In a used-car market there are good cars (worth €15,000 to the buyer, €12,000 to the seller) and bad cars (€7,000 to the buyer, €5,000 to the seller). The buyer cannot distinguish the types.

- (a) If the share of good cars is $p = 3/4$, what is the buyer's WTP?
- (b) Will sellers of good cars sell? And sellers of bad cars?
- (c) If instead the share is $p = 0$, what happens to the market?
- (d) What is the critical share p^* above which good cars are also traded?

Exercise 1—Solution

(a) $WTP = p \cdot 15,000 + (1 - p) \cdot 7,000 = \frac{3}{4} \cdot 15,000 + \frac{1}{4} \cdot 7,000 = 11,250 + 1,750 = 13,000.$

Exercise 1—Solution

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(b) Good car: the seller sells if $13,000 \geq 12,000$. Yes ✓.

Bad car: the seller sells if $13,000 \geq 5,000$. Yes ✓.

With $p = 3/4$ both types sell: the market works without collapsing.

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(c) With $p = 0$: $WTP = 7,000$. Sellers of bad cars sell if $7,000 \geq 5,000$. Yes. Only bad cars remain, traded at €7,000.

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Bad car: the seller sells if $13,000 \geq 5,000$. Yes ✓.

With $p = 3/4$ both types sell: the market works without collapsing.

(c) With $p = 0$: $WTP = 7,000$. Sellers of bad cars sell if $7,000 \geq 5,000$. Yes. Only bad cars remain, traded at €7,000.

(d) Sellers of good cars sell if $WTP \geq 12,000$:

$$p \cdot 15,000 + (1 - p) \cdot 7,000 \geq 12,000$$

$$7,000 + 8,000 p \geq 12,000 \Rightarrow p^* = \frac{5}{8} = 0.625$$

For $p \geq 5/8$: both types sell. For $p < 5/8$: only bad cars.

Exercise 2—Spence Signaling

Two types of workers. Type H : productivity €70,000/year. Type L : productivity €40,000/year. Education costs €15,000 for H and €35,000 for L .

- (a) Employers offer €70,000 to graduates and €40,000 to non-graduates. Check the IC conditions for both types.
- (b) Does a separating equilibrium exist?
- (c) Suppose the cost for L falls to €20,000. What happens to the separating equilibrium?

Exercise 2—Solution

(a)

$$IC_H : 70,000 - 15,000 = 55,000 > 40,000 \quad \checkmark \quad (\text{worth getting a degree})$$

$$IC_L : 70,000 - 35,000 = 35,000 < 40,000 \quad \checkmark \quad (\text{not worth mimicking})$$

Exercise 2—Solution

(a)

$$IC_H : 70,000 - 15,000 = 55,000 > 40,000 \quad \checkmark \quad (\text{worth getting a degree})$$

$$IC_L : 70,000 - 35,000 = 35,000 < 40,000 \quad \checkmark \quad (\text{not worth mimicking})$$

(b) **Yes**, the separating equilibrium exists: H gets a degree, L does not. Employers correctly associate graduates with H and non-graduates with L , and no one has an incentive to deviate.

Exercise 2—Solution

(a)

$$IC_H : 70,000 - 15,000 = 55,000 > 40,000 \quad \checkmark \quad (\text{worth getting a degree})$$

$$IC_L : 70,000 - 35,000 = 35,000 < 40,000 \quad \checkmark \quad (\text{not worth mimicking})$$

(b) **Yes**, the separating equilibrium exists: H gets a degree, L does not. Employers correctly associate graduates with H and non-graduates with L , and no one has an incentive to deviate.

(c) With a cost for L equal to €20,000:

$$IC_L : 70,000 - 20,000 = 50,000 > 40,000 \quad \Rightarrow \text{worth mimicking!}$$

The separating equilibrium **breaks down**: it would require $c_H < w_H - w_L < c_L$, i.e. $15,000 < 30,000 < 20,000$, which is impossible. If everyone gets a degree (*pooling*), the wage becomes average productivity $\bar{w}$: with pessimistic off-path beliefs (non-graduates are L), pooling holds only if $\bar{w} - 20,000 \geq 40,000$, i.e. with a share of H of at least $2/3$. The type composition and beliefs are therefore needed to conclude; in any case the degree no longer distinguishes the types.

Exercise 3—Managerial Moral Hazard

A firm hires a manager. Profit depends on the manager's effort:

- **High effort** (personal cost €50,000): profit €500,000 with prob. 0.8; €100,000 with prob. 0.2
- **Low effort** (cost €0): profit €500,000 with prob. 0.3; €100,000 with prob. 0.7

- (a) Compute the firm's expected profit with high and with low effort.
- (b) If the firm pays a fixed salary of €100,000, which effort does the manager choose?
- (c) Propose an incentive scheme that induces high effort.

Exercise 3—Solution

(a)

$$E[\pi \mid \text{high}] = 0.8 \cdot 500,000 + 0.2 \cdot 100,000 = 420,000$$

$$E[\pi \mid \text{low}] = 0.3 \cdot 500,000 + 0.7 \cdot 100,000 = 220,000$$

High effort raises expected profit by €200,000.

Exercise 3—Solution

(a)

$$E[\pi \mid \text{high}] = 0.8 \cdot 500,000 + 0.2 \cdot 100,000 = 420,000$$

$$E[\pi \mid \text{low}] = 0.3 \cdot 500,000 + 0.7 \cdot 100,000 = 220,000$$

High effort raises expected profit by €200,000.

(b) With a fixed salary, the manager's payoff is $100,000 - c$ where c is the cost of effort. High: $100,000 - 50,000 = 50,000$. Low: $100,000$. The manager chooses **low effort** (moral hazard).

Exercise 3—Solution

(a)

$$E[\pi \mid \text{high}] = 0.8 \cdot 500,000 + 0.2 \cdot 100,000 = 420,000$$

$$E[\pi \mid \text{low}] = 0.3 \cdot 500,000 + 0.7 \cdot 100,000 = 220,000$$

High effort raises expected profit by €200,000.

(b) With a fixed salary, the manager's payoff is $100,000 - c$ where c is the cost of effort. High: $100,000 - 50,000 = 50,000$. Low: $100,000$. The manager chooses **low effort** (moral hazard).

(c) *Performance-related bonus*. Example: base salary €80,000 + bonus €100,000 if profit = 500,000.

$$E[w \mid \text{high}] - 50,000 = 0.8 \cdot 180 + 0.2 \cdot 80 - 50 = 110 \text{ (in thousands of €)}$$

$$E[w \mid \text{low}] = 0.3 \cdot 180 + 0.7 \cdot 80 = 110$$

The manager is *indifferent*: a higher bonus (e.g. a contingent €120,000) is needed to strictly induce high effort.

General Summary (1/2)

Two types of asymmetric information

- **Hidden information** $\Rightarrow$ adverse selection: the worst types remain in the market
- **Hidden action** $\Rightarrow$ moral hazard: those who are covered take less precaution

Adverse selection (Akerlof)

- The average price discourages owners of high-quality goods
- Downward spiral: can lead to market collapse
- Examples: used cars, insurance, credit, the labor market

Moral hazard

- Those protected from risk manage it less well
- Examples: insurance, employment contracts, the banking system (*too big to fail*)

Comparisons

- Monopoly/oligopoly: DWL from market power ($P > MC$, Q too low)
- Asymmetric information: DWL from incomplete information (efficient trades do not take place)
- Both are market failures; both can justify public intervention

General Summary (2/2)

Private remedies

- **Signaling (Spence)**: the informed party reveals its type with a costly signal (e.g., education)
- **Screening**: the uninformed party designs self-selecting menus of contracts (e.g., insurance menus with different deductibles)
- **Reputation, warranties, certifications**: effective in repeated markets
- **Digital platforms**: aggregate dispersed information (AutoScout24, Airbnb, Amazon)
- **Telematics**: technological monitoring that reduces both adverse selection and moral hazard (black box in car insurance)

Public remedies

- Compulsory insurance, disclosure requirements, quality standards
- Skin in the game, capital requirements, legal liability

Policy lesson

Information asymmetries are among the most widespread and least visible market failures. They often justify public intervention, but regulation itself can create new moral hazard and adverse selection problems: a case-by-case analysis is always needed.

Skills Acquired

- ✓ Distinguish hidden information (adverse selection) from hidden action (moral hazard)
- ✓ Explain Akerlof's model and compute the critical threshold p^* for market collapse
- ✓ Apply the logic of adverse selection to insurance and the credit market
- ✓ Recognize moral hazard in insurance contracts and in principal-agent relationships
- ✓ Construct a separating equilibrium in Spence's signaling model (IC check)
- ✓ Distinguish signaling from screening, and identify which party moves first
- ✓ Evaluate regulatory tools against adverse selection and moral hazard

Optional Homework Exercises

1. Used-car market: good cars (€20,000 / €12,000) and bad cars (€8,000 / €5,000). With $p = 2/3$, does the market work? Find the critical share p^* .
2. An insurance company offers two policies: A (premium €200, deductible €800) and B (premium €500, deductible 0). Low-risk types have accident prob. 10%, high-risk types 40%. Damage: €1,000. Use the ICs to check that both types prefer the “right” policy.
3. A manager chooses between high effort (cost 80, prob. of success 90%) and low effort (cost 0, prob. 40%). Success is worth 1,000, failure 0. What is the socially optimal effort? With a fixed contract of 200, which effort does the manager choose?
4. Why can compulsory health insurance (such as the Italian SSN) solve the adverse selection problem that plagues private insurance? What new moral hazard problems can it create?
5. Explain why multi-year product warranties work as a credible signal of quality. Why can a low-quality producer not mimic this signal?